TRI-OCTAGON MATHEMATICAL RESEARCH

# Phase Reconstruction and Stationary Support Axes in the Tri-Octagon Map

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**Abstract**

On an accepted invariant coherent domain of the native Tri-Octagon map, five real retained coordinates determine the full complex state up to common phase. We recover the complete native history from their closed evolution and one initial common phase. A canonical representative gives an explicit unit multiplier and a continuous ordinary increment with absolute-summability control and the inherited limiting-phase bound. The adjacent-state overlap is retained-data dependent and nonvanishing, but its angle differs from the canonical increment by the angle of the inner product of consecutive representatives. We prove the branch-resolved correction, its accumulation rule and its change under a different representative; an analytic family shows that the correction is not identically zero. Composing reconstruction with the adopted lossless face decoder yields exact vector histories. Pure rephasing rotates each face vector about its fixed normal line, and those positioned axes coincide with the existing stationary support lines, with reversed directed-angle convention. In contrast, a single proper ambient rotation representing that action for every admitted state and every fixed channel label exists only for whole-turn phases. The complete native step can also change amplitudes and relative phases. We separate these results from staged observations, memory updates and observations from current memory, and from any assignment of a material feature or rigid-body pose.

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# 1 Question and scope

The retained coordinates of the coherent triad distinguish the component amplitudes and relative phases, but discard a common phase. Relative phase describes how the channels differ from one another. Common phase rotates all three complex components together while preserving their pair coherences. Once the original common phase is supplied, the existing native map determines its subsequent evolution; there is no additional phase law to choose.

This distinction becomes spatial through the adopted face decoder. Each complex channel supplies the two coefficients of an arrow in a fixed tangent plane. A common complex rephasing rotates these arrows in their respective planes. Whether the resulting axes coincide with existing stationary supports, and whether the three vector changes form one common proper ambient rotation, are precise questions about the adopted definitions. They do not require a material-motion interpretation.

The published predecessor establishes the retained coordinates, their exact image, autonomous evolution and energy retention [1]. We combine the accepted phase and spatial investigations [4, 5]: exact state reconstruction, the nontrivial overlap-angle correction, positioned-axis correspondence, and a quantified proper-rotation test. Their continuous-domain arguments are presented here. The inherited invariant-domain and convergence results [2], decoder [3], fixed-gap geometry [6], and observation definitions [7] retain their own hypotheses. No unrestricted coordinate-minimality or external priority claim is made.

# 2 Native setting and inherited retained dynamics

## 2.1 The ordered map and its regular domain

Let $\Omega \in \mathbb{C}^3$, and fix

$$\varepsilon = \frac{1}{10}, \quad g = \frac{1}{60}, \quad \lambda = \frac{1}{300}, \quad k = \left(\frac{999}{1000}, 1, \frac{1001}{1000}\right), \qquad L = \mathbf{1}\mathbf{1}^T - 3I. \tag{2.1}$$

With $s_i = |\Omega_i|^2$, the complex prestage and complete update are

$$M = I + \varepsilon\,\mathrm{diag}(k - s) + gL, \qquad \widetilde{\Omega} = M\Omega, \tag{2.2}$$

$$\delta_i = \lambda \sum_{j \neq i} \sin(3(\mathrm{Arg}_0\,\widetilde{\Omega}_j - \mathrm{Arg}_0\,\widetilde{\Omega}_i)), \qquad F(\Omega) = U_\delta \widetilde{\Omega}, \quad U_\delta = \mathrm{diag}(e^{i\delta_i}). \tag{2.3}$$

All kicks use the same prestage. The native convention is $\mathrm{Arg}_0(0) = 0$; at $\lambda = 0$, the synchronizer bypasses phase extraction and returns the prestage. Neither convention is modified here.

Set $D_0 = \pi/12$. The closed domain $\mathcal{D}$ has component amplitudes in $[9/10, 11/10]$ and ordinary phases contained in one interval of width at most $D_0$, with arbitrary common orientation. These are ordinary full-turn phases, not phases modulo $2\pi/3$. The following are inherited from [2, Sections 6–7] and restated in [1, Section 4]:

1. Positive prestage mixing preserves the input phase interval, and
$$a_* := \frac{88701}{100000} + \frac{3}{100}\cos D_0 \le |\widetilde{\Omega}_i| \le \frac{107701}{100000}, \qquad \frac{9}{10} < a_* \le |\widetilde{\Omega}_i| < \frac{11}{10}. \tag{2.4}$$
2. The simultaneous phase stage preserves that interval. The domain is forward invariant, with nonzero prestages at every iterate.
3. For consistent ordinary lifts and their enclosing intervals $I_n$,
$$I_{n+1} \subseteq I_n, \qquad \Delta_n := |I_n| \le c^n \Delta_{\mathrm{init}}, \qquad c = 1 - \frac{3\sqrt{2}}{200} \in (0, 1). \tag{2.5}$$

There is a unique positive aligned fixed profile $r_*$ in the amplitude cube, and

$$\Omega_n \longrightarrow e^{i\theta_\infty} r_*, \qquad |\theta_{i,n} - \theta_\infty| \le \Delta_{\text{init}} c^n. \tag{2.6}$$

We use these results without re-proving convergence. Regular common-phase equivariance is also inherited: at nonzero prestages, $F(u\Omega) = uF(\Omega)$ for $|u| = 1$. The harmonic sine is unchanged by the principal-argument wraps. The native zero stratum does not support an unrestricted extension of this assertion [3].

Table 1: Notation. The spatial support lines and native graph matrix are different objects.

| Symbol | Meaning |
|---|---|
| $\widetilde{\Omega},\ v_i$ | Complex prestage; decoded real face vector |
| $y,\ z(y),\ G$ | Retained tuple, canonical representative, closed retained update |
| $\Psi, \widehat{F}, \kappa$ | Full-state reconstruction, lifted update, unit phase multiplier |
| $A_{ij}, \eta_i, \vartheta$ | Signed pair areas, relative phases, canonical ordinary increment |
| $H,\ C_{\text{ch}}$ | Raw complex coherence; raw channel chirality |
| $n_i, t_i, \ell_i, L_k$ | Oriented face normal, tangent, positioned normal line, support line |
| $\theta_n, \theta_{\text{obs}}, t, n, \tau$ | Native anchor phase, observer angle, observer time, update index, static geometry parameter |
| $N_\Omega$ | Observer norm $\sqrt{\sum_i s_i}$, never the multiplier $\kappa$ |

## 2.2 Representative, exact image and closed map

Retain the source convention

$$\overline{\Omega}_i \Omega_j = S_{ij} + iA_{ij}, \qquad H = \Omega\Omega^\dagger = S_G - iA_G, \qquad y = \mathcal{R}(\Omega) = (s_1, s_2, s_3, A_{12}, A_{13}). \tag{2.7}$$

The inherited reconstruction and fibre theorem [1, Section 5] gives

$$b_2 = +\sqrt{s_1 s_2 - A_{12}^2}, \qquad b_3 = +\sqrt{s_1 s_3 - A_{13}^2},$$

$$z(y) = \left(\sqrt{s_1}, \frac{b_2 + iA_{12}}{\sqrt{s_1}}, \frac{b_3 + iA_{13}}{\sqrt{s_1}}\right)^T, \qquad H(y) = z(y)z(y)^\dagger, \tag{2.8}$$

$$S_{23} = \frac{b_2 b_3 + A_{12}A_{13}}{s_1}, \qquad A_{23} = \frac{b_2 A_{13} - A_{12} b_3}{s_1}. \tag{2.9}$$

Its fibres are exactly common-phase orbits. In particular $H_{12} = b_2 - iA_{12}$, not $b_2 + iA_{12}$.

The exact image $\mathcal{Y} = \mathcal{R}(\mathcal{D})$ consists of

$$\frac{81}{100} \le s_i \le \frac{121}{100}, \qquad \eta_1 = 0, \quad \eta_2 = \arcsin\frac{A_{12}}{\sqrt{s_1 s_2}}, \quad \eta_3 = \arcsin\frac{A_{13}}{\sqrt{s_1 s_3}},$$

$$\Delta(y) := \max(0, \eta_2, \eta_3) - \min(0, \eta_2, \eta_3) \le D_0, \tag{2.10}$$

where the arcsines are real principal values. They equal $\operatorname{Arg} z_i(y)$. Separate area bounds would not impose the required joint span. The radicands in (2.8) are at least $(81/100)^2 \cos^2 D_0 > 0$, and $s_1 \ge 81/100$. Thus closed boundaries, coincident phases, equal amplitudes and zero individual areas are regular; no area is a denominator.

For completeness the inherited retained map can be written entirely in these data:

$$\widetilde{H} = M(y)H(y)M(y)^T, \qquad \rho_i = \sqrt{\widetilde{H}_{ii}} > 0,$$

$$\delta_i(y) = \lambda \sum_{j \ne i} \operatorname{Im}\left[\left(\frac{\widetilde{H}_{ji}}{\rho_j \rho_i}\right)^3\right], \qquad H^+ = U_\delta \widetilde{H} U_\delta^\dagger, \tag{2.11}$$

$$Gy = (H_{11}^+, H_{22}^+, H_{33}^+, -\operatorname{Im} H_{12}^+, -\operatorname{Im} H_{13}^+).$$

The $ji$ index and extraction minus signs preserve (2.7). We use the accepted statements $\mathcal{R}F = G\mathcal{R}$ and $G\mathcal{Y} \subseteq \mathcal{Y}$, including all subsequent mathematical iterates. No coordinate-minimality result is required below.

## 3 Exact reconstruction of the native phase

**Theorem 3.1** (Full-state reconstruction)**.** *The map $\Psi : \mathcal{Y} \times \mathrm{U}(1) \to \mathcal{D}$, $\Psi(y,u) = uz(y)$, is a homeomorphism with inverse*

$$\Psi^{-1}(\Omega) = \left(\mathcal{R}(\Omega), \frac{\Omega_1}{|\Omega_1|}\right). \tag{3.1}$$

*Define*

$$\kappa(y) = \frac{F_1(z(y))}{|F_1(z(y))|} = e^{i\delta_1(y)} \frac{(M(y)H(y))_{11}}{\sqrt{s_1}\rho_1(y)}. \tag{3.2}$$

*Then $F(z(y)) = \kappa(y)z(Gy)$, and the lifted update*

$$\widehat{F}(y,u) = (Gy, u\kappa(y)) \quad \textit{satisfies} \quad F\Psi = \Psi\widehat{F}. \tag{3.3}$$

*For $u_0 = \Omega_{1,0}/|\Omega_{1,0}|$, every admitted iterate is*

$$\boxed{\Omega_n = u_0 \left(\prod_{j=0}^{n-1} \kappa(G^j y_0)\right) z(G^n y_0), \qquad n \geq 0.} \tag{3.4}$$

*Proof.* The inherited fibre theorem and positive real first component of $z$ give (3.1), proving bijectivity. The formulas and positive denominators show continuity in both directions. Since $H = zz^\dagger$,

$$(MH)_{11} = (Mz)_1 \overline{z}_1 = \sqrt{s_1}(Mz)_1, \qquad |F_1(z)| = \rho_1 > 0$$

by (2.4); this proves (3.2). The states $Fz(y)$ and $z(Gy)$ have the same retained tuple, so differ by one common unit scalar. Their first components identify it as $\kappa(y)$. Regular equivariance gives (3.3) and induction gives (3.4). Forward invariance keeps every step regular. The empty product is one. □

This is conjugacy of self-maps, requiring no invertibility of the dynamics. If only $y_0$ is given, (3.4) is the family indexed by $u_0 \in \mathrm{U}(1)$; invariant data do not determine the discarded initial common phase.

**Theorem 3.2** (Canonical ordinary increment and accumulation)**.** *On the entire closed image $\mathcal{Y}$, the principal argument*

$$\vartheta(y) = \operatorname{Arg} F_1(z(y)) \tag{3.5}$$

*is continuous, satisfies $\kappa = e^{i\vartheta}$, and obeys $|\vartheta(y)| \leq \Delta(y)$. For a chosen real lift $e^{i\theta_0} = u_0$, set*

$$\theta_{n+1} = \theta_n + \vartheta(y_n), \qquad y_n = G^n y_0. \tag{3.6}$$

*Then the full-state relation is $\Omega_n = e^{i\theta_n} z(y_n)$. Its limiting phase is*

$$\theta_\infty = \theta_0 + \sum_{j=0}^{\infty} \vartheta(y_j), \tag{3.7}$$

*with the distinct bounds*

$$\sum_{j \geq n} |\vartheta(y_j)| \leq \frac{\Delta_{\mathrm{init}} c^n}{1-c}, \qquad |\theta_\infty - \theta_n| \leq \Delta_{\mathrm{init}} c^n. \tag{3.8}$$

*Proof.* Put $m = \min_i \eta_i(y)$, $M_a = \max_i \eta_i(y)$. The canonical interval $[m, M_a]$ contains zero and has width $\Delta \le D_0$. Inherited prestage mixing gives lifts $\psi_i \in [m, M_a]$; inherited synchronization gives output lifts $\beta_i = \psi_i + \delta_i$ in the same interval. Hence

$$\operatorname{Re} F_1(z(y)) = \rho_1 \cos\beta_1 \ge a_* \cos\Delta \ge a_* \cos D_0 > 0. \tag{3.9}$$

The continuous right-half-plane argument is therefore a single global branch, and $\vartheta = \beta_1 \in [m, M_a]$. Nonvanishing alone would not justify that branch. In particular $\vartheta = \psi_1 + \delta_1$: it includes the prestage phase change and is not generally only the kick.

The output relative phases are $\beta_i - \beta_1 \in [-D_0, D_0]$. The principal arcsines in (2.10) recover these differences exactly, so

$$\eta_i(Gy) = \beta_i - \beta_1, \qquad \theta_{i,n} = \theta_n + \eta_i(y_n) \tag{3.10}$$

propagates the inherited consistent ordinary lift. There is no later unresolved multiple of $2\pi$. Now $|\vartheta(y_n)| \le \Delta_n \le c^n \Delta_{\text{init}}$ proves the absolute-series bound and (3.7). The same lift identifies the limit with the inherited $\theta_\infty$. Both $\theta_n$ and that limiting phase lie in $I_n$, giving the stronger net bound $|\theta_\infty - \theta_n| \le \Delta_n$, hence the second part of (3.8). □

The absolute variation and net remaining adjustment in (3.8) are not interchangeable. The sharper net estimate is inherited phase control, not a new optimal-rate or smooth-asymptotic-phase theorem. Changing $\theta_0$ by $2\pi\ell$ changes all real lifts equally and leaves the complex history unchanged.

On aligned input, $A_{12} = A_{13} = 0$, so $z(y) = r > 0$. The accepted aligned amplitude map is

$$\mathcal{A}_i(r) = [1 - 2g + \varepsilon(k_i - r_i^2)]r_i + g(r_j + r_k) > 0.$$

The kicks vanish, $\kappa = 1$ and $\vartheta = 0$, while $\mathcal{A}(r)$ may differ from $r$. The amplitudes converge toward $r_*$. Zero phase increment therefore does not mean fixed full state or zero potential decrease.

## 4 Overlap angle and representative correction

The adjacent-state overlap is the scalar $\mathcal{O}(\Omega) = \Omega^\dagger F(\Omega)$, distinct from the placed-Z observer. Its conjugation order is essential.

**Theorem 4.1** (A branch-resolved overlap correction)**.** *The overlap depends only on retained data and satisfies*

$$\mathcal{O}(y) = \operatorname{tr}(U_\delta M H(y)) = \sum_i e^{i\delta_i}(MH(y))_{ii} = \kappa(y) B_{\text{sec}}(y), \qquad B_{\text{sec}}(y) = z(y)^\dagger z(Gy). \tag{4.1}$$

*Its principal argument $\alpha_{\text{ov}} = \operatorname{Arg}\mathcal{O}$ and $\gamma = \operatorname{Arg} B_{\text{sec}}$ are continuous on $\mathcal{Y}$. With $\Delta = \Delta(y)$,*

$$\operatorname{Re}\mathcal{O}(y) \ge \frac{243}{100}\cos\Delta \ge \frac{243}{100}\cos D_0 > 0, \qquad |\alpha_{\text{ov}}| \le \Delta, \tag{4.2}$$

$$\operatorname{Re} B_{\text{sec}}(y) \ge \frac{243}{100}\cos(2\Delta) \ge \frac{243\sqrt{3}}{200} > 0, \qquad |\gamma| \le 2\Delta. \tag{4.3}$$

*The real principal values obey the exact equality*

$$\boxed{\alpha_{\text{ov}}(y) = \vartheta(y) + \gamma(y).} \tag{4.4}$$

*Proof.* Using $H = \Omega\Omega^\dagger$ and trace cyclicity gives $\Omega^\dagger U_\delta M \Omega = \mathrm{tr}(U_\delta M H)$. All factors are determined by (2.11). Equivariance also directly proves common-rephasing invariance of this overlap. Applying Theorem 3.1 to the canonical representative gives the last equality in (4.1).

The inherited input and output phase lifts lie in the same interval of width $\Delta$. Thus, writing $r_i = |\Omega_i|$,

$$\mathcal{O} = \sum_i r_i \rho_i e^{i d_i}, \qquad d_i = \theta_i^+ - \theta_i \in [-\Delta, \Delta].$$

Each weight is positive, and their sum is at least $3(9/10)^2 = 243/100$. This proves the real-part bound in (4.2). The sector $|\arg w| \le \Delta < \pi/2$ is convex: it is given by $|\operatorname{Im} w| \le \tan\Delta \operatorname{Re} w$, $\operatorname{Re} w \ge 0$. Adding the positive-weight terms gives the angle bound as well.

For the canonical representatives, $|\eta_i(y)| \le \Delta$ and $|\eta_i(Gy)| \le \Delta(Gy) \le \Delta$. Hence

$$B_{\mathrm{sec}} = \sum_i r_i \rho_i e^{i(\eta_i(Gy) - \eta_i(y))}, \qquad |\eta_i(Gy) - \eta_i(y)| \le 2\Delta \le \pi/6.$$

The same sector argument proves (4.3). Both scalar functions are continuous and uniformly inside the open right half-plane, giving the claimed principal branches. Factorization first gives $\alpha_{\mathrm{ov}} \equiv \vartheta + \gamma \pmod{2\pi}$; but $|\vartheta + \gamma| \le 3\Delta \le \pi/4 < \pi$. Thus the principal values agree as real numbers, proving (4.4). □

Consequently reconstructing the anchor phase by overlap requires

$$\theta_N - \theta_0 = \sum_{j=0}^{N-1} \alpha_{\mathrm{ov}}(y_j) - \sum_{j=0}^{N-1} \gamma(y_j). \tag{4.5}$$

Both series converge absolutely, with respective absolute tails at most $\Delta_{\mathrm{init}} c^n/(1-c)$ and $2\Delta_{\mathrm{init}} c^n/(1-c)$. Their difference has the sharper net-tail estimate in (3.8). That estimate is not separately assigned to the uncorrected overlap sum. On the aligned subset all three angles vanish, even during amplitude relaxation.

The correction is not identically zero. Appendix A analytically differentiates the native map on $z(t) = (1, e^{it}, e^{it})$ at the fixed coefficients and obtains $\gamma'(0) < 0$. This is distinct from a numerical native-step evaluation. It establishes a local difference of the two angles, without classifying their agreement set. We neither assume that $\gamma$ is an endpoint difference nor prove that such a representation is impossible.

## 4.1 Changing the representative

For a specified $h : \mathcal{Y} \to \mathrm{U}(1)$, let $z_h(y) = h(y) z(y)$. Equivariance gives

$$\kappa_h(y) = \frac{h(y)}{h(Gy)} \kappa(y), \qquad B_{\mathrm{sec},h}(y) = \overline{h(y)} h(Gy) B_{\mathrm{sec}}(y). \tag{4.6}$$

Indeed $F z_h(y) = h(y)\kappa(y) z(Gy) = \kappa_h(y) z_h(Gy)$, while the second expression is the inner product of the new representatives. Their factors cancel in $\mathcal{O} = \kappa_h B_{\mathrm{sec},h}$. For $h_j = h(y_j)$, finite cancellation gives

$$\prod_{j<N} \kappa_h(y_j) = \frac{h_0}{h_N} \prod_{j<N} \kappa(y_j), \qquad \prod_{j<N} B_{\mathrm{sec},h}(y_j) = \frac{h_N}{h_0} \prod_{j<N} B_{\mathrm{sec}}(y_j). \tag{4.7}$$

Taking $u_{h,0} = u_0/h_0$ gives $u_{h,N} = u_N/h_N$ and

$$u_{h,N} z_h(y_N) = u_N z(y_N) = \Omega_N. \tag{4.8}$$

These identities hold at the unit-complex level without selecting a real argument of $h$.

If a real lift $h = e^{i\xi}$ has been specified, the corresponding real increment lifts are

$$\vartheta_h^{\mathrm{lift}} = \vartheta + \xi(y) - \xi(Gy), \qquad \gamma_h^{\mathrm{lift}} = \gamma - \xi(y) + \xi(Gy), \tag{4.9}$$
$$e^{i\vartheta_h^{\mathrm{lift}}} = \kappa_h, \qquad e^{i\gamma_h^{\mathrm{lift}}} = \frac{B_{\mathrm{sec},h}}{|B_{\mathrm{sec},h}|}.$$

Their principal arguments are only congruent to these real lifts unless further branch bounds apply. The two increment lifts sum pointwise to $\alpha_{\mathrm{ov}}$. Their finite sums acquire opposite endpoint terms $\xi(y_0) - \xi(y_N)$, which cancel. For example, $h = e^{-i\eta_2}$ is a specified continuous channel-2 anchor with real lift $-\eta_2$. Its reconstruction increment is $\vartheta + \eta_2(Gy) - \eta_2(y)$, the ordinary increment of channel 2.

Arbitrary continuous representative changes need not retain the canonical pointwise increment bounds or absolute-variation bounds. Even if $h(y_n)$ converges, continuity alone does not control the total variation of its successive values. These endpoint changes of convention do not establish an endpoint formula for the original correction $\gamma$, and do not supply geometric holonomy.

# 5 Decoded phase and positioned support axes

## 5.1 The adopted frame

The existing shell decoder uses $e_z = (0, 0, 1)$, $r = \sqrt{3}/6$ and $O = (0, r, 0)$. Table 2 transcribes the defining NRG frame table [3, Appendix, notation and conventions] together with the reference/PLUS support convention of [6, Sections 1 and 3]. Native labels $1, 2, 3$ correspond to faces A=P1, B=P2, C=P3; the implementation uses indices $0, 1, 2$.

Table 2: Exact frame and support dictionary in the common right-handed ambient frame. Coordinates are triples; every displayed final coordinate is zero.

| Channel / face | $c_i$ | $n_i$ | $t_i = e_z \times n_i$ | $g_{i-1}$ |
|---|---|---|---|---|
| 1 / A=P1 | $(-\frac{1}{4}, \frac{\sqrt{3}}{4}, 0)$ | $(-\frac{\sqrt{3}}{2}, \frac{1}{2}, 0)$ | $(-\frac{1}{2}, -\frac{\sqrt{3}}{2}, 0)$ | $(\frac{\sqrt{3}}{2}, -\frac{1}{2}, 0)$ |
| 2 / B=P2 | $(0, 0, 0)$ | $(0, -1, 0)$ | $(1, 0, 0)$ | $(0, 1, 0)$ |
| 3 / C=P3 | $(\frac{1}{4}, \frac{\sqrt{3}}{4}, 0)$ | $(\frac{\sqrt{3}}{2}, \frac{1}{2}, 0)$ | $(-\frac{1}{2}, \frac{\sqrt{3}}{2}, 0)$ | $(-\frac{\sqrt{3}}{2}, -\frac{1}{2}, 0)$ |

The fixed-gap source uses centered coordinates $(X, Y, w)$, with $e_w = e_z$, about precisely this uncentered $O$. Its directions are

$$e_j = (\cos(-\pi/2 + j\pi/3), \sin(-\pi/2 + j\pi/3), 0), \qquad g_k = e_{2k+1}.$$

Thus its centered support $\mathbb{R}g_k$ is the positioned line $L_k = O + \mathbb{R}g_k$. The planar source operator $J(X, Y) = (-Y, X)$ has the orientation of $e_z\times$. No reflection, fit or channel permutation is used. The fixed-gap source's other symbol $n_k = e_{2k}$ denotes hexagon directions, not the decoder normals.

**Proposition 5.1** (Positioned axes and directed signs)**.** *For the defining frames and support directions,*

$$\boxed{n_i = -g_{i-1}, \qquad c_i - O = r n_i, \qquad \ell_i := c_i + \mathbb{R}n_i = O + \mathbb{R}g_{i-1} = L_{i-1}.} \tag{5.1}$$

*For right-hand rotations, $R_{n_i}(\chi) = R_{g_{i-1}}(-\chi)$. Their affine actions satisfy, for every ambient point $X$,*

$$c_i + R_{n_i}(\chi)(X - c_i) = O + R_{g_{i-1}}(-\chi)(X - O). \tag{5.2}$$

*Proof.* The table gives opposite normals and support directions. Subtracting $O$ from the three centers gives

$$(-1/4, \sqrt{3}/12, 0), \quad (0, -\sqrt{3}/6, 0), \quad (1/4, \sqrt{3}/12, 0),$$

which are exactly $rn_i$. Hence $c_i + \mathbb{R}n_i = O + (r + \mathbb{R})n_i = O + \mathbb{R}g_{i-1}$. Rodrigues' formula

$$R_n(\chi) = \cos\chi\, I + \sin\chi\, [n]_\times + (1 - \cos\chi) nn^T, \qquad [n]_\times v = n \times v,$$

gives the signed-angle identity. Since $c_i - O$ lies on that axis, it is fixed by the rotation; expanding both sides gives (5.2). □

These are equal positioned, unoriented lines, not merely parallel directions. They are three different lines through $O$, not a single common axis. Their accepted stationarity does not identify a material rod or specify what is attached to them [6].

## 5.2 Lossless vectors and pure phase action

Put $W_i = n_i^\perp$. The adopted decoder and encoder are

$$D_i(q + ip) = qt_i + pe_z, \qquad E_i(v) = t_i \cdot v + ie_z \cdot v. \tag{5.3}$$

The frame is orthonormal with

$$n_i \times t_i = e_z, \qquad n_i \times e_z = -t_i. \tag{5.4}$$

Thus $E_i D_i = I$, $\|D_i\omega\| = |\omega|$, and $D_i E_i = I - n_i n_i^T$ on ambient vectors. In particular $D_i E_i = I$ on $W_i$. The combined maps $D, E$ are inverse between $\mathbb{C}^3$ and $\bigoplus_i W_i$, not between arbitrary ambient arrays [3, State–shell attachment, `eq:decoder`].

**Theorem 5.2** (Spatial action of common phase)**.** *For all complex $\omega$ and real $\chi$,*

$$\boxed{D_i(e^{i\chi}\omega) = \cos\chi\, D_i\omega + \sin\chi\, n_i \times D_i\omega = R_{n_i}(\chi) D_i\omega = R_{g_{i-1}}(-\chi) D_i\omega.} \tag{5.5}$$

*The endpoint based at $c_i$ rotates about the positioned line $\ell_i = L_{i-1}$.*

*Proof.* For $\omega = q + ip$, the real and imaginary parts after multiplication are $q\cos\chi - p\sin\chi$ and $q\sin\chi + p\cos\chi$. Substitution in (5.3) and (5.4) proves the first identity. The normal term in Rodrigues' formula vanishes by tangency; Proposition 5.1 gives the last equality. For $X = c_i + D_i\omega$, the new endpoint is $c_i + R_{n_i}(\chi)(X - c_i)$, so (5.2) identifies its positioned axis. □

The algebra includes $\omega = 0$; this does not extend the native reconstruction theorem to singular prestages or other coefficient histories. These based vectors lie in unbounded tangent planes. Their endpoints need not lie in the finite source faces, and are not automatically particle positions or material features.

## 5.3 The complete decoded update

The accepted matched transport is

$$T_{ij} = t_i t_j^T + e_z e_z^T : W_j \to W_i, \qquad T_{ij} D_j\omega = D_i\omega.$$

It is an isometry on its tangent domain and rank two ambiently. The existing conjugate face update is $F_{\text{face}} = DFE$, explicitly

$$\begin{aligned} \widetilde{v}_i &= v_i + \varepsilon(k_i - \|v_i\|^2) v_i + g \sum_{j \neq i} (T_{ij} v_j - v_i), \\ v_i^+ &= \cos\delta_i\, \widetilde{v}_i + \sin\delta_i\, n_i \times \widetilde{v}_i. \end{aligned} \tag{5.6}$$

The kicks are those of the encoded complex prestage, with the original conventions [3, `eq:faceupdate`]. No second dynamics or carrier motion is added.

Define $\mathcal{D}_{\mathrm{face}}(y,u) = D(uz(y))$. Then $ED = I$ and Theorem 3.1 give

$$F_{\mathrm{face}}\mathcal{D}_{\mathrm{face}}(y,u) = DF(uz(y)) = D(u\kappa(y)z(Gy)) = \mathcal{D}_{\mathrm{face}}\widehat{F}(y,u). \tag{5.7}$$

This proves the commuting relation, and composition with (3.4) yields the exact history

$$v_{i,n} = D_i\left[u_0\left(\prod_{j=0}^{n-1}\kappa(G^j y_0)\right) z_i(G^n y_0)\right]. \tag{5.8}$$

Losslessness is preserved: encoding these vectors recovers $\Omega_n$, then (3.1) recovers $(y_n, u_n)$.

Pure rephasing changes $u$ at fixed $y$. An actual native step can change $y$ as well. Using the accepted ordinary lifts (3.10),

$$\theta_{i,n+1} - \theta_{i,n} = \vartheta(y_n) + \eta_i(Gy_n) - \eta_i(y_n), \tag{5.9}$$

as an equality of real lifts. The vector has length $\sqrt{s_{i,n}}$ and direction $\cos\theta_{i,n}\, t_i + \sin\theta_{i,n}\, e_z$. Consequently the full decoded step is

$$\boxed{v_{i,n+1} = \sqrt{\frac{s_{i,n+1}}{s_{i,n}}}\, R_{n_i}[\vartheta(y_n) + \eta_i(Gy_n) - \eta_i(y_n)]\, v_{i,n}.} \tag{5.10}$$

This combines amplitude change with anchor and relative-phase change; it is not a pure common rotation or a continuous interpolating path. Aligned amplitudes can relax while every angular increment vanishes. Closed-boundary and zero-area cases cause no singularity because $s_{i,n} \geq 81/100$.

Finally, the representative change of (4.8) gives $D_i(u_h z_{h,i}) = D_i(uz_i)$. Thus the decoded vectors are independent of that bookkeeping convention, although $\vartheta$ is not. Changing a representative together with its inverse phase factor differs from actually rephasing the native state.

## 6 When one proper rotation represents all three vectors

The three local rotations in Theorem 5.2 need not be one common ambient rotation. The precise universal question fixes the labels and quantifies over all admitted states.

**Theorem 6.1** (Universal proper-rotation classification)**.** *For fixed $\chi \in \mathbb{R}$, there exists a single $Q_\chi \in SO(3)$, independent of $\Omega$, such that for every $\Omega \in \mathcal{D}$ and every fixed channel label,*

$$D_i(e^{i\chi}\Omega_i) = Q_\chi D_i(\Omega_i), \qquad i = 1,2,3, \tag{6.1}$$

*if and only if $\chi \equiv 0 \pmod{2\pi}$.*

*Proof.* First take $\Omega = (1,1,1)$, with unit moduli and phase width zero. Its decoded vectors are $t_i$; after rephasing they are $w_i = \cos\chi\, t_i + \sin\chi\, e_z$. For $i \neq j$,

$$t_i \cdot t_j = -\frac{1}{2}, \qquad w_i \cdot w_j = -\frac{1}{2}\cos^2\chi + \sin^2\chi = -\frac{1}{2} + \frac{3}{2}\sin^2\chi. \tag{6.2}$$

Orthogonal transformations preserve these inner products, forcing $\sin\chi = 0$. Only 0 or $\pi$ modulo $2\pi$ remain.

For the half-turn, take $\Omega = e^{i\pi/4}(1,1,1)$. It is also admitted: $D_0$ bounds the phase width, not the absolute common angle. Its decoded vector columns form

$$V = \frac{1}{\sqrt{2}} \begin{pmatrix} -1/2 & 1 & -1/2 \\ -\sqrt{3}/2 & 0 & \sqrt{3}/2 \\ 1 & 1 & 1 \end{pmatrix}, \qquad \det V = \frac{3\sqrt{6}}{8} > 0. \tag{6.3}$$

At $\chi = \pi$ each column is negated. Hence $Q_\pi V = -V$, and full rank forces $Q_\pi = -I$, whose determinant is $-1$. It is not proper. Conversely, for $\chi \equiv 0 \pmod{2\pi}$, all complex components are unchanged and $Q_\chi = I$ works for every state. □

The witnesses are exact symbolic states, not native-step experiments. No parameters are changed. The first triple is planar: at $\chi = \pi$, the proper rotation $R_{e_z}(\pi)$ does negate its three tangents. The full-rank second witness is what excludes that half-turn universally. Thus the theorem does not say that every individual pair of vector triples lacks a proper registration.

This comparison concerns free vectors in a fixed ambient frame. It does not move centers, permute channels, carry the mesh, establish a native period or classify individual registrations. Neither this obstruction nor a successful registration of one triple identifies the pose of an attached central object. The fixed-gap support stabilizers concern specified common rigid transformations and attachment conditions [6, Section 3]; they answer a different question.

# 7 Observations, energy consistency and interpretation

## 7.1 Retained observations and phase-sensitive memory

The existing raw channel quantities are

$$I(y) = s_1 + s_2 + s_3, \qquad N_\Omega = \sqrt{I}, \qquad C_{\mathrm{ch}}(y) = (A_{23}, -A_{13}, A_{12}), \tag{7.1}$$

where $A_{23}$ is supplied by (2.9). For $\Omega = q + ip$, $C_{\mathrm{ch}} = q \times p$. It is not the distinct ambient axial projection $W = T_f C_{\mathrm{ch}}$, with $T_f = [t_1\ t_2\ t_3]$ [3]. The historical staged Z uses the raw $C_{\mathrm{ch}}$.

With the specified external observer and placement settings, staged Z has the form [7, Section 3]

$$\begin{aligned} K(N_\Omega) &= \lambda_{\mathrm{vp}} \frac{N_\Omega}{1 + N_\Omega}, \quad E_{\mathrm{obs}}(t) = e^{-\gamma_{\mathrm{obs}} t}, \\ H_{\theta_{\mathrm{obs}}} &= \cos[3(\theta_{\mathrm{obs}} - \ell)](\cos\theta_{\mathrm{obs}}, \sin\theta_{\mathrm{obs}}, 1), \\ Z &= \alpha K E_{\mathrm{obs}} H_{\theta_{\mathrm{obs}}} + \beta C_{\mathrm{ch}}, \qquad X = O + \mu R_\psi Z. \end{aligned} \tag{7.2}$$

These coefficients and clocks are external observer data. No new preset is selected. If $\sigma_n$ collects those data, then $Z_n = \mathcal{Z}(y_n; \sigma_n)$ and $X_n = \mathcal{X}(y_n; \sigma_n)$. Equal retained starts and identical exogenous schedules give equal staged and placed histories, since $y_n = G^n y_0$. Their face-vector histories may differ through $u_0$. This is composition of accepted closure with observation invariance, not an autonomous recurrence on Z alone. Frozen settings still leave dependence on changing $I(y)$ and $C_{\mathrm{ch}}(y)$, so the point can move.

The separate EMA innovation is

$$J_n = \mathrm{Im}\left[u_n z_1(y_n)\overline{z_2(y_n)} z_3(y_n)\right], \qquad j_n = \frac{J_n}{1 + |J_n|}. \tag{7.3}$$

The cubic product has common-phase weight one: substituting $\Omega = uz$ yields $u\overline{u}u = u$. The source `advance_ema` uses $j_n$, current memory and literal innovation weight 0.01. Generating

memory needs the initial memory and specified sampling/update order in addition to a native history, or instead a supplied innovation sequence [8].

In contrast, `observe_ema` observes *current* memory without evaluating the cubic or updating memory. Its scalar is the norm/clock harmonic plus current memory, followed by the existing macro direction and raw-chirality blend. Thus its value is determined by $I(y_n)$, $C_{\mathrm{ch}}(y_n)$, current memory and observer configuration/clock, with placement data added for a point. At fixed values of these inputs, current common phase supplies no additional value information. Phase can affect later observations through explicitly updated memory. These are distinct operations, not two descriptions of one implicit update [7, Section 8][8].

Table 3: Observation dependencies. History inputs include the specified native or observer ordering; no canonical schedule is inferred.

| Quantity | Inputs for a current value | Additional history information and phase dependence |
|---|---|---|
| Face arrows | $y_n, u_n$, fixed frames | $y_0, u_0$ determine their native history through (5.8). |
| Intensity and raw chirality | $y_n$ | $y_0, G$ suffice; no common-phase value is needed. |
| Staged Z / placed point | $y_n$, observer data; placement for the point | Equal $y_0$ and schedules give equal histories. No recurrence on Z alone is claimed. |
| EMA innovation | $y_n, u_n$, via (7.3) | Reconstructed state determines the phase-sensitive cubic. |
| EMA memory update | Normalized innovation and current memory | Requires $y_0, u_0$, initial memory, existing rule and sampling/order; a supplied innovation sequence can replace the native-history inputs. |
| EMA observation from current memory | $I(y_n), C_{\mathrm{ch}}(y_n)$, memory, clock/configuration; placement if requested | At fixed listed inputs, current phase adds no value information. Later values can depend on phase through explicit memory updates. Generating memory requires the preceding row's inputs. |

No unknown past memory is recovered. Native common phase, observer angle, observer time, update index and static shape parameter $\tau$ remain distinct. Historical Z01 used $(\varepsilon, g, \lambda) = (.05, .2, .001)$, $k = (1, 1, 1)$; Z02/Z03 used $(1/10, 1/5, 17/100)$, $k = (2, 3, 4)$ [7, Section 2]. The present domain theorem is not transferred to those histories, and none is replayed here.

## 7.2 Energy consistency

The predecessor retains the edge-once potential

$$\mathcal{V}(\Omega) = \varepsilon \sum_i \left( \frac{|\Omega_i|^4}{4} - \frac{k_i |\Omega_i|^2}{2} \right) + \frac{g}{2} \sum_{i<j} |\Omega_i - \Omega_j|^2$$

and every matched edge energy $E_{ij} = s_i + s_j - 2S_{ij}$ [1, Sections 4 and 6]. Multiplication by a common unit scalar leaves each distance, intensity and the norm of $\widetilde{\Omega} - \Omega$ unchanged. The same isometry under the matched decoder preserves those individual energies, independent of representative choice. Thus the inherited bound remains

$$\mathcal{V}(F\Omega) - \mathcal{V}(\Omega) \le -\frac{16869}{20000} \|\widetilde{\Omega} - \Omega\|_2^2, \qquad \Omega \in \mathcal{D}. \tag{7.4}$$

Equality in monotonicity occurs exactly on the accepted aligned fixed circle. The coefficient multiplies the squared prestage increment, not a full-step distance or percentage of potential. This consistency statement supplies no physical energy or flux identification and requires no new descent proof.

### 7.3 What the correspondence establishes

One initial common phase completes the retained state history. The overlap diagnostic encodes a different angle and requires its representative correction. The adopted decoder then supplies exact arrows and mathematical endpoints: their pure-phase rotation axes are the positioned stationary support lines, with the directed sign fixed by the source frames. An actual native step can additionally change their lengths and relative phases, and a nontrivial universal common proper rotation of the free-vector triple is excluded by Theorem 6.1.

Identifying a named material feature with an arrow or prescribing its registration to a stationary line would add a declared correspondence. Neither a moving support, filled aperture, hinge, continuous interpolating path, flow, force nor emission law follows from these results. The central broken opening arcs and filled host-notch measurement patches remain distinct. No feature must be chosen to review this paper.

The close original enclosing sphere is a fixed geometric reference; the separately discussed tight toroid remains a distinct intended feature. Two co-present arrangements and rapid alternation remain different alternatives. The manuscript selects none of them. Native/core motion and gravity–magnetism remain linked research directions, with outer structure and justified SRG integration later; TORMENT remains separate and protected, and Three-Way remains parked. These standing directions are retained, not activated by this write-up.

**Attribution and verification scope.** The two underlying proof reviews have project-internal acceptance [4, 5]. Their existing exact-algebra records accompany this research edition unchanged. The present work assembles the proofs, incorporates the approved editorial and observer-table corrections, and verifies the compiled document. It adds no scientific execution, literature/novelty investigation or external peer-review claim. GPT project-internal scientific and manuscript review is complete; the author approved repository publication on 2 November 2026. This is the author-authorized repository research edition v0.1.

## A An analytic witness for the overlap correction

We retain the local sign witness of [4, Section 5.3] to show that $\gamma$ in Theorem 4.1 is not identically zero. This calculation analytically differentiates the native map. It is not a numerical native-step evaluation, sampled trajectory, parameter scan or new continuation problem.

Take the canonical family $z(t) = (1, e^{it}, e^{it})$, $|t| \le D_0$. Its amplitudes are one and its phase width is $|t|$, so every input is admitted. The intensities and hence $M$ are constant on this family. At $t = 0$, its prestage is the positive vector

$$w_i = 1 + \varepsilon(k_i - 1), \qquad w = \left(\frac{9999}{10000}, 1, \frac{10001}{10000}\right), \qquad \sum_i w_i = 3. \tag{A.1}$$

Differentiating the prestage at zero gives $i(2g, w_2 - g, w_3 - g)$, so its phase derivatives are

$$\tau_1 = \frac{2g}{w_1}, \qquad \tau_2 = 1 - \frac{g}{w_2}, \qquad \tau_3 = 1 - \frac{g}{w_3}. \tag{A.2}$$

Write $d = 1 - 9\lambda > 0$, $T = \sum_i \tau_i$. Differentiating the simultaneous kicks at the aligned prestage gives output phase derivatives

$$\sigma_i = \tau_i + 3\lambda \sum_{j \neq i} (\tau_j - \tau_i) = d\tau_i + 3\lambda T. \tag{A.3}$$

The output modulus derivatives vanish: the baseline vector is real positive and its derivative is imaginary. Removing the output channel-1 phase therefore gives derivatives $iw_i(\sigma_i - \sigma_1)$ for the next canonical representative. The current conjugate input derivatives are $(0, -i, -i)$. Thus

$$B_{\mathrm{sec}}(0) = 3, \qquad \operatorname{Im} B'_{\mathrm{sec}}(0) = \sum_i w_i \sigma_i - 3\sigma_1 - (w_2 + w_3). \tag{A.4}$$

Since $\sum_i w_i \tau_i = w_2 + w_3$, substitution cancels the $T$ terms and yields

$$\operatorname{Im} B'_{\mathrm{sec}}(0) = -9\lambda(w_2 + w_3) - 3d\tau_1, \qquad \boxed{\gamma'(0) = -3\lambda(w_2 + w_3) - d\tau_1 < 0.} \tag{A.5}$$

The established right-half-plane branch justifies differentiating its argument and dividing by $B_{\mathrm{sec}}(0) = 3$. Therefore $\gamma(t) \neq 0$ for sufficiently small positive $t$, and $\alpha_{\mathrm{ov}} \neq \vartheta$ there. The conclusion is local nonidentity, not a classification of all states where the angles agree or a verdict on endpoint-difference representations.

## References


[1] H. F. Halldórsson. *Native Potential Descent and Closed Coherence Coordinates in the Tri-Octagon Map*. Repository research edition v0.1, 2 November 2026. Published reading edition, commit `634ff13b6666`. Sections 4–6 supply the fixed domain, reconstruction and energy retention.

[2] H. F. Halldórsson. *Transient Collective Turning and Coherent Relaxation in the Native Triad*. Repository research edition v0.1, 28 October 2026. Sections 6–7, especially `lem:phase`, `thm:convergence`, `eq:nesting` and `eq:phasebound`. Accepted reading edition; compact source snapshot accompanies this package.

[3] Tri-Octagon research programme. *Native Response Geometry and Spectral Circulation in the Tri-Octagon Map*. Research publication v0.1, 6 October 2026. State–shell attachment, `eq:decoder`, `eq:faceupdate`; notation appendix's exact shell-frame table. Transport and closure source; frame appendix.

[4] Tri-Octagon research programme. *Native common-phase reconstruction: proof review*. Task `NATIVE_PHASE_RECONSTRUCTION_01`, 2 November 2026. Packaged proof note and existing exact-algebra record. Unpublished project research evidence; project-internal acceptance recorded in the manuscript preparation order.

[5] Tri-Octagon research programme. *Native phase and the existing spatial decoder*. Task `NATIVE_SPATIAL_CORRESPONDENCE_01`, 2 November 2026. Packaged proof note, claim ledger and existing exact-algebra record. Unpublished project research evidence; project-internal acceptance and observer-table correction recorded in the manuscript preparation order.

[6] Tri-Octagon research programme. *NCM-FIXED-GAP-01 v0.1*. External geometry investigation, 31 October 2026; accepted GPT review of the same date. Sections 1 and 3 fix the common origin, reference support directions and attachment distinction. Packaged accepted report; acceptance review. Unpublished project research evidence.

[7] H. F. Halldórsson. *Understanding Z Motion in the Tri-Octagon Model: Native changes, observer effects, and fixed-geometry source accounting*. Explanatory methods note, draft v0.1, 1 November 2026. Sections 2, 3 and 8. Repository reading copy; source snapshot accompanies this package.

[8] Tri-Octagon research programme. *Z manifold observation implementation*, `kernel_physics/z_manifold.py`. `cubic_j`, `normalized_cubic`, `advance_ema`, `ema_scalar`, `observe_ema`. Immutable reading reference; read-only source corroboration, not executed in this manuscript build.